# 1 Method Description

## 1.1 Overview

This simulator couples the finite-volume fluid-flow solver presented in Simulator 07 with the finite-element mechanical solver presented in Simulator 08. The governing equations and numerical discretizations of the individual solvers are unchanged. The purpose of the coupled simulator is therefore to account for the two-way interaction between pore pressure and mechanical deformation.

The flow model computes the pore-pressure field, which is passed to the mechanical solver through Biot's effective stress principle. The resulting deformation produces a volumetric strain field, which modifies the pore volume and therefore the porosity used by the flow solver. This coupling is performed iteratively at every time step until convergence.

## 1.2 Flow Equation

The fluid-flow model solves the slightly compressible single-phase flow equation,

$$\frac{\partial(\phi\rho)}{\partial t} = \nabla \cdot \left(\frac{\rho k}{\mu}\nabla P\right) + \frac{Q}{V}, \tag{1}$$

where $P$ is pore pressure, $\rho$ is fluid density, $\phi$ is porosity, $k$ is permeability, $\mu$ is viscosity, $V$ is the cell volume, and $Q$ represents sources and sinks. The finite-volume discretization and nonlinear solution procedure are identical to those presented in Simulator 07.

## 1.3 Mechanical Equation

The mechanical response is obtained by solving the quasi-static equilibrium equation

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{f} = \mathbf{0}, \tag{2}$$

where $\mathbf{f}$ is the body force per unit volume. In porous media, total stress is written as

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}' + \alpha P\mathbf{I}, \tag{3}$$

where $\boldsymbol{\sigma}'$ is the effective stress, $\alpha$ is the Biot coefficient, and $\mathbf{I}$ is the identity tensor. The equilibrium equation therefore becomes

$$\nabla \cdot \left(\boldsymbol{\sigma}' + \alpha P\mathbf{I}\right) + \mathbf{f} = \mathbf{0}. \tag{4}$$

The effective stress is related to strain through linear elasticity,

$$\boldsymbol{\sigma}' = \mathbf{D}\boldsymbol{\epsilon}, \tag{5}$$

where $\mathbf{D}$ is the elastic constitutive matrix. The finite-element discretization, element assembly, and solution procedure are identical to those presented in Simulator 08.

### 1.4 Volumetric Strain

Once the displacement field has been obtained, the volumetric strain is calculated as

$$\epsilon_v = \epsilon_{xx} + \epsilon_{yy}, \tag{6}$$

or equivalently,

$$\epsilon_v = \mathbf{m}^T \mathbf{B}_e \mathbf{u}_e, \tag{7}$$

where

$$\mathbf{m} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}. \tag{8}$$

The volumetric strain provides the coupling between the mechanical and flow models by modifying the pore volume.

## 2 Coupling

The mechanical model returns the volumetric strain, which is used to update the porosity in the flow model according to

$$\phi^0 = \phi_0 + (1 - \phi_0)\epsilon_v, \tag{9}$$

where $\phi^0$ is iteration baseline porosity, $\phi_0$ is the reference porosity, and $\epsilon_v$ is the volumetric strain. Note that porosity is a function of $\phi^0$ and $P$.

The flow and mechanical simulators are coupled through a sequential fixed-point iteration performed at every time step. At the beginning of each time step, the converged state from the previous time step, $\Psi_0$, is taken as the initial estimate. The flow model is solved first, producing an updated state $\Psi$ containing the pressure, fluid density, porosity, and other flow variables. The resulting pore-pressure field is then passed to the mechanical model, which computes the corresponding displacement and volumetric strain fields.

The updated volumetric strain modifies the iteration baseline porosity, $\phi^0$, and, by extension, the actual porosity, $\phi = \phi^0 e^{c_\phi (P - P_0)}$. Since this change alters the pore volume without any associated fluid transport, the pressure must be adjusted to conserve fluid mass before the next flow iteration. For each control volume, the fluid mass immediately before the mechanical solve is

$$m_0 = V\rho\phi, \tag{10}$$

where $V$ is the cell volume. After updating the porosity from the computed strain, the pressure is corrected so that this mass is preserved. This is achieved by solving the nonlinear equation

$$r(P) = V\rho(P)\phi(P) - m_0 = 0, \tag{11}$$

using Newton's method. The Jacobian is

$$J(P) = V\left(\phi\frac{\partial\rho}{\partial P} + \rho\frac{\partial\phi}{\partial P}\right), \tag{12}$$

and the pressure correction is

$$\Delta P = -\frac{r}{J}. \tag{13}$$

The density and porosity are updated after each Newton iteration until the mass residual satisfies the prescribed tolerance. The corrected pressure field then serves as the initial pressure estimate for the subsequent flow solve.

Following the mechanical solve, the pressure field is compared with that from the previous coupling iteration. If the pressure change exceeds the convergence tolerance, another flow–mechanics iteration is performed. This procedure continues until both the flow and mechanical models have converged, after which the simulator advances to the next time step.

At each time step, the coupled simulator proceeds as follows:

1. Initialise the coupling iteration using the converged state from the previous time step.
2. Solve the fluid-flow equations.
3. Store the output state from the flow model.
4. Solve the mechanical equilibrium equations using the updated pore pressure.
5. Compute the volumetric strain.
6. Update the baseline porosity.
7. Perform the local pressure correction to conserve fluid mass with respect to the stored state in step 3.
8. If the pressure change exceeds the convergence tolerance, return to Step 2.
9. Advance to the next time step.

Note finally that an additional formulation for porosity is also sometimes used based on Coussy (2007),

$$\phi = -\alpha\epsilon_v + \frac{P(\alpha - \phi_0)}{K_s} + \phi_0, \tag{14}$$

where $K_s$ is the grain bulk modulus.

# Bibliography

Coussy, O. (2007). Revisiting the constitutive equations of unsaturated porous solids using a Lagrangian saturation concept. International Journal for Numerical and Analytical Methods in Geomechanics, 31:1675–1694. https://doi.org/10.1002/nag.613.